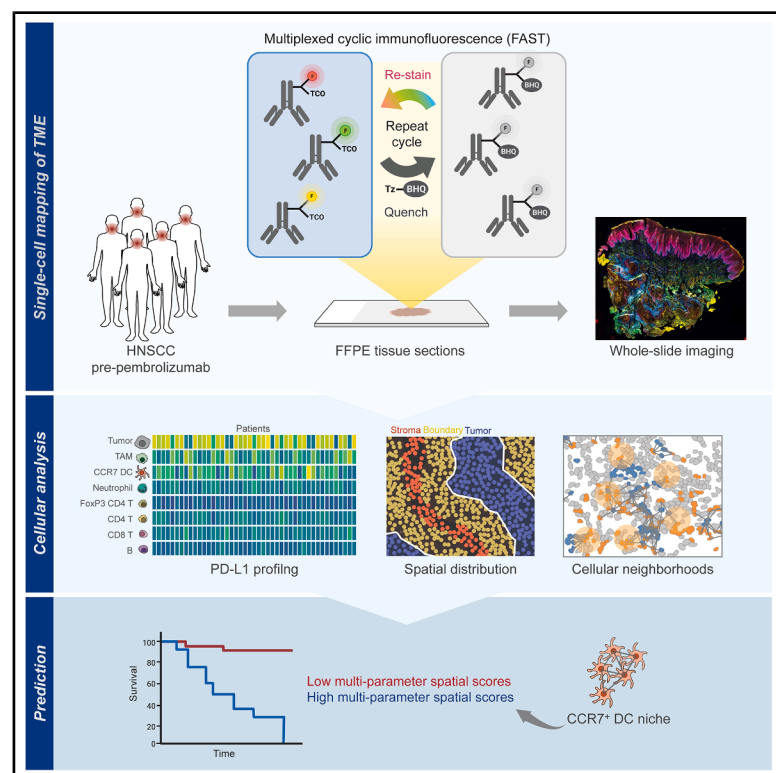


Spatial analysis identifies DC niches as predictors of pembrolizumab therapy in head and neck squamous cell cancer

Graphical abstract



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In brief

The treatment of head and neck squamous cell carcinoma remains a significant health challenge. Multiplexed spatial analysis of the tumor microenvironment reveals the limitations of current PD-L1 scoring in predicting treatment response. Instead, multiparametric spatial scores, which incorporate dendritic cell niche profiling, improve immunotherapy prediction.

Highlights

- PD-L1 expression correlates poorly with immunotherapy response in HNSCC
- FAST cyclic imaging quantifies expression of 21 protein markers in 41 HNSCC patient samples
- The spatial organization of CCR7⁺ DCs in niches predicts survival to immunotherapy
- MPSSs outperform single-feature analyses in prediction of clinical outcomes



Article

Spatial analysis identifies DC niches as predictors of pembrolizumab therapy in head and neck squamous cell cancer

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SUMMARY

Head and neck squamous cell carcinoma (HNSCC) shows variable response to anti-programmed cell death protein 1 (PD-1) therapy, which can be partially explained by a combined positive score (CPS) of tumor and immune cell expression of programmed death-ligand 1 (PD-L1) within the local tumor microenvironment (TME). To better define TME immune determinants associated with treatment efficacy, we conduct a study of $n = 48$ HNSCC tumors from patients prior to pembrolizumab therapy. Our investigation combines a rapid bioorthogonal multiplex staining method with computational analysis of whole-slide imaging to capture the single-cell spatial heterogeneity and complexity of the TME. Analyzing 6,316 fields of view (FOVs), we provide comprehensive PD-L1 phenotyping and cell proximity assays across the entirety of tissue sections. While none of the PD-L1 metrics adequately predict response, we find that the spatial organization of CCR7⁺ dendritic cells (DCs) in niches better predicts overall patient survival than CPS alone. This study highlights the importance of understanding the spatial context of immune networks for immunotherapy.

INTRODUCTION

Head and neck squamous cell carcinoma (HNSCC) is a worldwide public health problem, and until recently, patients with HNSCC have had limited effective treatment options. In 2019, the Food and Drug Administration (FDA) approved using anti-PD-1 monotherapy and its combination with chemotherapy as first-line treatments for patients with recurrent and/or metastatic HNSCC. In part, because PD-L1 expression is typically associated with response to anti-PD-1 monotherapy, the FDA requires a minimum combined positive score (CPS) of 1 in tumor tissues for treatment eligibility.¹ This marked the first approval of a companion diagnostic test in HNSCC care. Despite this paradigm

shift to implementing a histopathological biomarker, challenges persist. Most importantly, while 85% of HNSCC express PD-L1, only 10%–20% of patients respond to anti-PD-1 monotherapy¹ and only ~30% to anti-PD-1 + chemotherapy. This discrepancy is not unique to HNSCC, as recent studies have also pointed out the limitations of relying solely on PD-L1 as a diagnostic marker in other cancers, including non-small cell lung cancer,² triple-negative breast cancer,³ and urothelial carcinoma.⁴ Consequently, there is a pressing unmet clinical need to identify better biomarkers of response and, correspondingly, innovative technologies that facilitate multi-cellular phenotypic and functional analysis of the dynamic tumor microenvironment (TME) within short turnaround times, as not to delay clinical treatment.



Addressing these needs ranks among the top challenges in advancing cancer immunotherapy.⁵

To objectively determine the characteristics of HNSCC, we analyzed the tissue sections in their entirety from 48 patients diagnosed with HNSCCs. All patients underwent tissue collection prior to the start of anti-PD-1 immunotherapy and were followed longitudinally. Using a recently developed multiplexed spatial biology approach FAST⁶ (Figure S1), we interrogated the entire specimen landscape of all patients. This resulted in a rich dataset of 6,316 microscopic fields of view (FOVs) covering the entirety of all tissue samples, each with 21 distinct protein markers, allowing for cellular classification. Analysis was facilitated by an automated pipeline developed specifically for this study, which allowed the examination of nearly 3 million cells and hundreds of tissue parameters. The study was designed to address three main questions: (1) what is the accuracy of the CPS metric as assessed by the standard of care tissue processing in HNSCC?¹; (2) is there potential for enhanced PD-L1 profiling metrics to refine predictions of immunotherapy response?; and (3) are there other biomarkers or cellular configurations that could provide better predictive accuracy? For the latter objective, we selected 21 biomarkers based on our current understanding of immune responses in HNSCC.⁷ Our results show that spatial analysis of tumor-immune and immune-immune cell interactions introduces new metrics (multiparametric spatial scores, MPSSs) with high predictability of response. Many of the most predictive metrics involved LAMP3⁺ dendritic cell (DC) subtype 3 (termed CCR7⁺ DCs from here on)^{8,9} and their interactions with surrounding cells. These metrics, which integrate the diverse aspects of the TME into our understanding of tumor biology, hold potential for further exploration in diagnostics and prediction.

RESULTS

Multiplexed spatial single-cell analysis pipeline identifies highly heterogeneous TME landscapes in patients with HNSCC

Multiplexed single-cell quantification of protein expression in the entire TME in HNSCC patient samples was achieved utilizing a cyclic imaging approach (FAST)^{6,10} (Figure 1). We used 7 cycles of three-channel immunofluorescence (plus nuclear staining in the fourth channel) with minimal spectral overlap between the fluorophores (Table S1). As a result, 21 protein markers were imaged across the entire specimen typically comprising hundreds of FOVs per sample ($561.6 \times 561.6 \mu\text{m}$ of imaging area for each FOV with 10% overlap between FOVs for raster scanning). The entire tumor area along with the adjacent resident tissue area was imaged to capture the heterogeneity of the TME within each patient sample. Each marker (Table S1) was validated against positive and negative controls and against conventional immunohistochemistry (IHC) and immunofluorescence using human tonsil sections and HNSCC tissue arrays (Figures S1B and S1C). A total of 48 patient samples with a wide range of clinical characteristics, spanning diverse primary tumor sites and the clinical spectrum from primary tumors to locoregional recurrences to distant metastases, were obtained prior to anti-PD-1 treatment (pembrolizumab) and analyzed (Figure S2). Of these,

$n = 7$ samples did not pass all the quality control steps (Figure S3A), resulting in the final analysis of $n = 41$ patients with a total of 6,316 different FOVs with nearly 3 million segmented cells covering the whole tumor areas and their immediately adjacent resident tissue areas (Table S4). This study population included 14 patients with complete or partial responses, 2 patients with mixed responses, and 25 patients with progressive or stable disease (Figure S2).

Figure 2 illustrates the 7-cycle FAST imaging of a representative HNSCC sample (Figures 2A and 2B) and how such multiplexed image data were processed to obtain cell phenotypes based on protein marker annotation (Figures 2C and Figure S1). Individually segmented cells were assigned to specific cell types as presented in the synthetic images of representative FOVs in Figure 2D. The density of different immune and tumor cells, their PD-L1 expression, and their spatial distribution within the entire tumor tissue were further analyzed to generate different tissue scores (Figures 2E and Figure S3B). This example also demonstrates immune and tumor heterogeneity within tumor tissue (Figure 2D). Together, this analysis allowed us to convert the multiplexed raw datasets into quantitative maps of cell types and their activation states and evaluate intratumoral heterogeneity.

We first performed a broad survey of tumor and immune composition across all patient samples (Figures 3A–3D). This revealed a considerable range of immune, host, and tumor cell heterogeneity with broad overlap among patients regardless of response to anti-PD-1 treatment. While samples of “inflamed” phenotypes were found in the responder group, more responders had “immune desert,” “immune excluded,” or “cold” phenotypes, suggesting that such classification is not sufficient to distinguish responders prior to anti-PD-1 treatment (Figure 3C).^{11–13}

Subsequently, we compared three broad classes of cellular metrics (Figure 3E), beginning with the abundance parameters of single-cell types previously shown to be associated with clinical outcomes or responses to immunotherapy in other cancers. Such examples include tumor-infiltrating lymphocytes (TILs)¹⁴ or specific types of myeloid cells (tumor-associated macrophages, DCs, and neutrophils) in HNSCC.⁷ From this analysis, it emerged that no single immune cell frequency or density parameter showed a clear difference between responders and non-responders in our dataset, as determined by the area under the receiver operating characteristic (ROC) curve (AUC, Figure 3E). AUC values ≥ 0.8 were deemed good discrimination between classes.

Next, we tested whether subsets of immune cell populations with different proliferation and functional states would better discriminate between responders and non-responders. We therefore quantitated the expression levels of Ki-67 in all cell types, PD-1 in lymphocytes, CD163 in macrophages and other myeloid cells, and T-cell factor 1 (TCF1) in T cells (Figure 3E).^{15,16} Again, there were no appreciable differences in these numbers between responders and non-responders. Finally, we tested different ratios between two different cell populations that have been reported previously to indicate immunostimulatory or immunosuppressive TME (e.g., CD8⁺ T cell/FoxP3⁺CD4⁺ T cell ratio¹⁷ and immune cell/tumor cell ratio¹⁸). Among all the

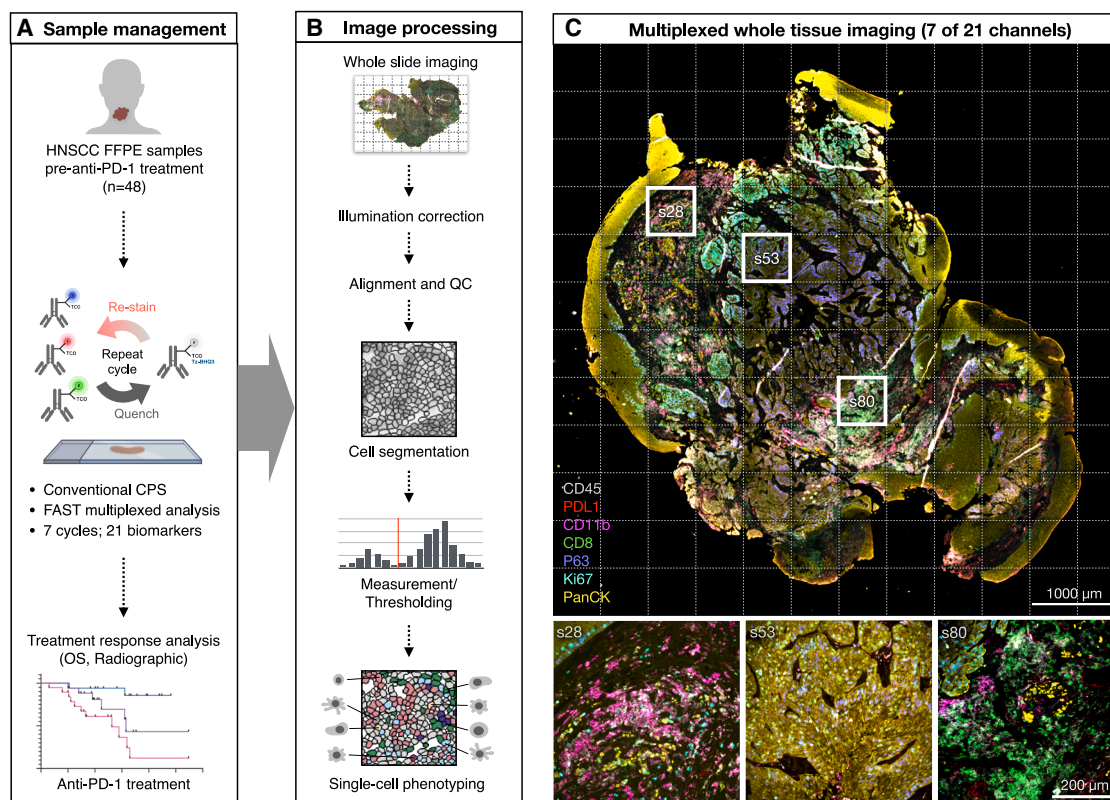


Figure 1. Study overview and sample processing

(A) Sample management. The cyclic immunostaining and whole-tissue scanning were performed on tissue specimens obtained from 48 patients prior to the initiation of anti-PD-1 immunotherapy. Treatment response was prospectively determined by advanced imaging, clinical information, and overall survival (OS). (B) Image processing. Images obtained from each field of view (FOV; $n = 6,316$ total for scanning the entire tumor tissues from all patient samples analyzed), 4 channels (one of which is always DAPI for image registration), and 7 image cycles were corrected for illumination and alignment. Subsequently, cells were segmented, and the fluorescence signal of each biomarker was quantified in individual cells. (C) Multiplexed whole-slide imaging of a representative tissue section (patient PIO50) over 144 separate FOVs to scan the entire specimen and 7 image cycles to measure 21 protein markers (scale bar: 1,000 μ m). White: CD45, red: PD-L1, magenta: CD11b, green: CD8, blue: P63, cyan: Ki-67, yellow: Pan-CK. Note the considerable heterogeneity across the tissue. The three composite images at the bottom are higher magnification images of example FOVs showing the marked heterogeneity (scale bar: 200 μ m). Imaging was done with a 20 $\times$ objective.

cellular metrics tested, the CD8⁺ T cell/FoxP3⁺CD4⁺ T cell ratio showed the highest AUC of 0.7.

PD-L1 profiling shows a poor correlation with immunotherapy response

One of the primary goals of the study was to determine whether PD-L1 profiling on specific cell types would be more predictive than the current gross CPS metric (Figure 4A). Determination of PD-L1 CPS (or tumor proportion score, TPS) involves the use of validated anti-PD-L1 antibodies, the performance of chemically amplified IHC on tissue sections,¹ and the assignment of a score based on cumulative PD-L1 positivity as a proportion of viable tumor cells. Prior research has shown that amplified IHC can produce distinct results depending on the antibodies and methods used.^{19–21} The clinical work-up herein used the E1L3N antibody (Cell Signaling Technologies) for scoring. As part of this research, we embedded PD-L1 detection within the broader profiling capabilities of multiplexed cyclic immunofluorescence, allowing us to separately and unequivocally

enumerate PD-L1 positivity by tumor cells and/or other cell types. Stringent validation experiments were performed to determine the specificity of the immunostaining and the precision of the analytical pipeline (Figure S1). At the initial global level, a direct comparison between clinical IHC-CPS and FAST-CPS from patient samples in this study showed a good correlation, yielding an R^2 value of 0.83 (Figures 4B and 4C). Irrespective of the scoring method, CPS had limited predictive value in response to anti-PD-1 therapy in HNSCC. While CPS was higher in the complete response group, there was considerable spread among patient groups (Figure 4D). The AUC for clinical IHC-CPS in differentiating responders from non-responders of this study was 0.52 and that of FAST-CPS was 0.58 (Figure 4F).

Using the FAST platform analysis, we next conducted a comprehensive profiling of PD-L1 expression across all cell types within the TME (Figure S4). We also determined whether PD-L1 expression varied among cells in different spatial compartments of the TME (e.g., within tumor nests, normal resident

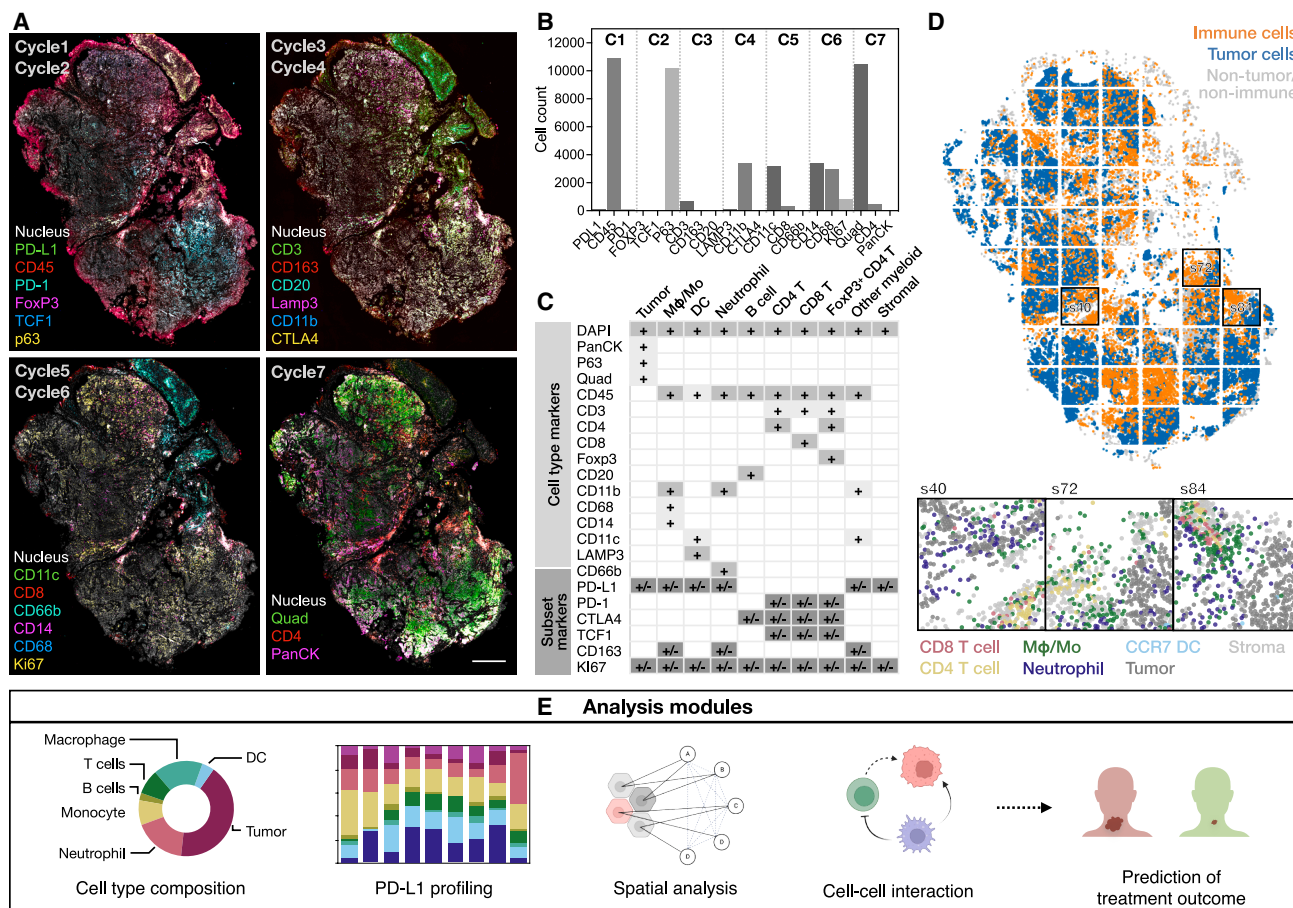


Figure 2. Data processing for the analysis of the HNSCC tumor microenvironment

(A) A representative example of multiple cycles of whole-tissue scanning with multi-color immunofluorescence, showing distinct staining patterns through cycles 1–7. Cycle 1: PD-L1 (green), CD45 (red), and PD-1 (cyan); cycle 2: FoxP3 (magenta), TCF1 (blue), and P63 (yellow); cycle 3: CD3 (green), CD163 (red), and CD20 (cyan); cycle 4: Lamp3 (magenta), CD11b (blue), and CTLA4 (yellow); cycle 5: CD11c (green), CD8 (red), and CD66b (cyan); cycle 6: CD14 (magenta), CD68 (blue), and Ki-67 (yellow); and cycle 7: Quad (green), CD4 (red), Pan-CK (magenta), and nucleus (white). In this example, each tissue scanning contains a total of 99 FOVs (scale bar: 500 μ m). See also [Tables S1](#) and [S2](#).

(B) The graph summarizes the cell counts expressing a specific marker in the example from (A).

(C) Protein marker combinations for identifying different cell types and their subtypes by antibody staining. “+”: markers for defining cell types; “+/–”: subset markers; mid gray boxes: required biomarker; light gray boxes: elective biomarkers, at least one required in case there are multiple of them; dark gray boxes: functional subset markers. Cells that are negative for immune and tumor cell markers include endothelial cells, fibroblasts, and others in the TME.

(D) Synthetic image derived from (A)–(C) where tumor cells are shown in blue, all immune cells (CD45⁺) in orange, and stromal cells in gray. The image on the left shows the distribution of immune cells across the entire specimen, while the images of higher resolution maps on the right show the distribution of different immune cell types in three representative FOVs (s40, s72, and s84).

(E) The overall downstream analysis involved determining (1) the density of different cell types ([Figure 3](#)), (2) PD-L1 expression in all cell types ([Figure 4](#)), and (3) the spatial distribution of different cell types in the tumoral and stromal area and their relative location to one another ([Figure 5](#)) in each sample.

See also [Figure S3B](#).

tissue areas, and tumor boundaries, [Figure S5A](#)). Our data showed that PD-L1 could be detected in various cell types, including tumor cells, myeloid cells, and others ([Figure 4E](#)). The highest expression was found in tumor cells (1.6 ± 0.8 -fold higher as compared to other cell types) and LAMP3⁺ DCs (2.5 ± 1.2 -fold higher as compared to other cell types) as determined by mean fluorescence intensity in individual cells. LAMP3⁺ DCs²² were previously also identified as DC3²³ and mregDC.²⁴ More recent comprehensive reviews and efforts toward standardizing nomenclature now refer to these cells as CCR7⁺ DCs.^{8,9} Given

the high expression level of PD-L1 at single-cell level and/or high fraction of PD-L1-expressing cells among tumor and myeloid cells (CCR7⁺ DCs, macrophages/monocytes), we concluded that these myeloid cell populations are the primary source of PD-L1 in HNSCC ([Figure S4](#)). However, further analysis of immune subtypes did not reveal patterns that could more accurately predict treatment outcomes. Specifically, the cellular PD-L1 levels among different cell types did not allow response classification. We saw no evidence that PD-L1 profiling of CCR7⁺ DCs (AUC 0.53 by proportion of CCR7⁺ DCs among all

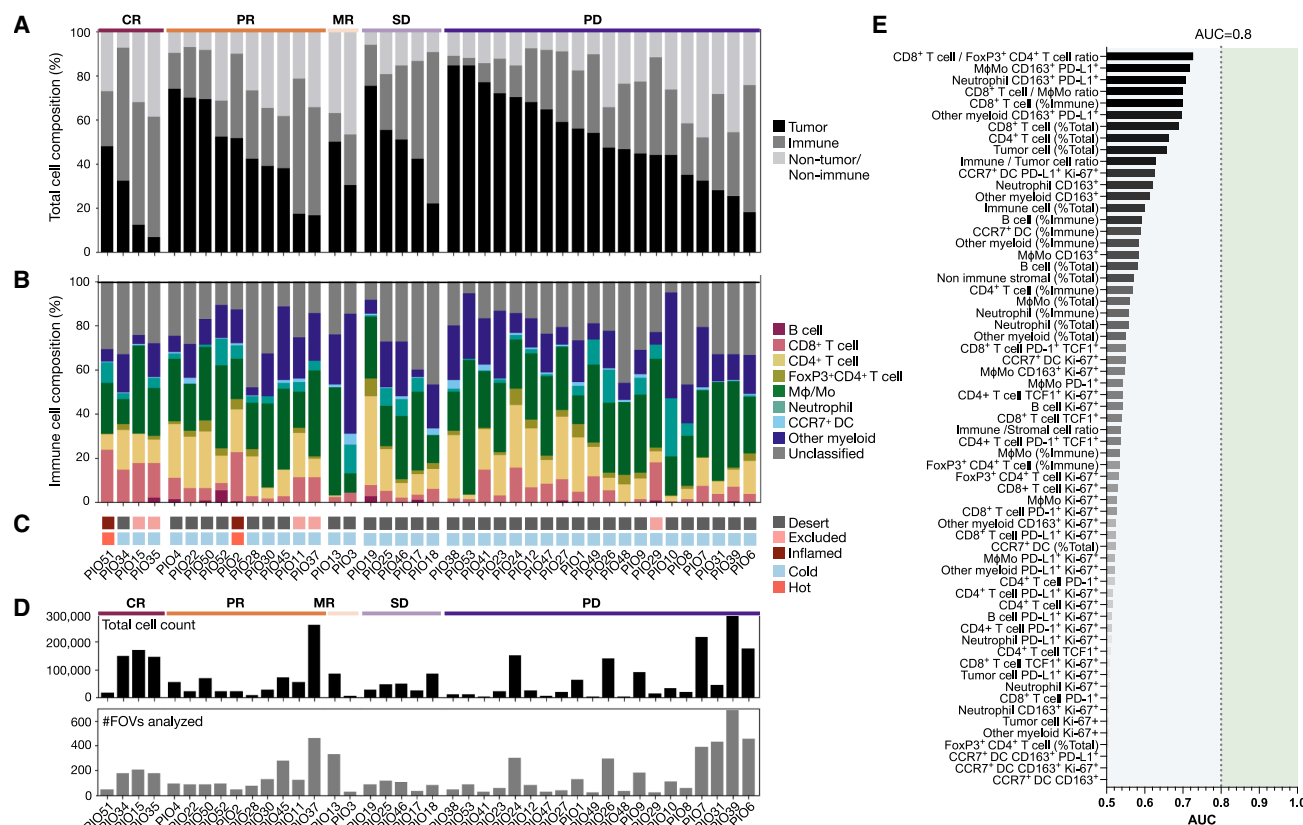


Figure 3. Immune profiling of HNSCC samples

Summary graphs showing cell composition across all FOVs analyzed in all samples that passed the quality control ($n = 41$). Non-responders to anti-PD-1 ($n = 29$) include patients with progressive disease (PD), stable disease (SD), or mixed response (MR). Responders ($n = 12$) to anti-PD-1 treatment including patients with complete response (CR) or partial response (PR). See Figure S2 and Table S3 for details on patient sample information.

(A) Cells identified in each sample were classified into the immune, tumor, and stromal cells, and their composition in each sample is demonstrated.

(B) Fraction of different immune cell types in each sample is exhibited. Unclassified cells represent CD45⁺ cells that could not be unequivocally assigned to one of the categories.

(C) The immune context of each sample was classified as “desert,” “excluded,” “inflamed,” “hot,” or “cold” based on the number of CD8⁺ T cell infiltration per unit tissue area.

(D) Summary of the total number of FOVs and cells analyzed for each patient. A total of 6,316 FOVs were analyzed, resulting in an average of 154 FOVs and 67,157 cells analyzed per sample.

(E) Areas under the ROC curve (AUC) of different immune cell metrics to show their performance in differentiating responders from non-responders. None of the metrics was able to achieve an AUC cutoff of 0.8.

PD-L1⁺ cells, 0.59 by fraction of PD-L1-expressing cells among CCR7⁺ DCs) or macrophages (AUC 0.54 by proportion of macrophages/monocytes among all PD-L1⁺ cells, 0.56 by fraction of PD-L1-expressing cells among macrophages/monocytes) between responders and non-responders reached an AUC of acceptable discrimination (Figure 4F). Figure 4F compares the AUC of all PD-L1 expression metrics tested, all of which were below 0.7. Comparing PD-L1 expression in diverse cell types within different spatial compartments of the TME also did not result in better predictions (Figure S8). We also tested the same list of parameters shown in Figures 3 and 4 among patients who received pembrolizumab monotherapy (excluding combination therapy group) to determine whether potential inter-patient heterogeneity caused by combination therapy affected the AUC values; however, no meaningful difference was observed (Figures S5B and S5C).

Spatial parameters as indicators of immunotherapy response

Given the weak predictive power of the aforementioned metrics, we next set out to examine whether the spatial distribution of diverse immune cell types or their proximity to one another would serve as better predictors (irrespective of PD-L1 expression). To achieve this, we defined tumor boundaries in the HNSCC tissues and analyzed the abundance of immune cell populations located in specific tissue compartments: tumor, normal resident tissue, and the boundary region (within 50 μm of the tumor boundary, Figure 5A). Interestingly, the abundance of CCR7⁺ DCs within the resident tissue regions showed greatest difference between responders and non-responders (log2 fold change 2.7; p value < 0.01 ; AUC 0.65; q value 0.38), underscoring the potential importance of these cells in shaping the TME to enhance responsiveness to anti-PD-1 therapy. The abundance of CD8⁺ T cells in tumor nests was

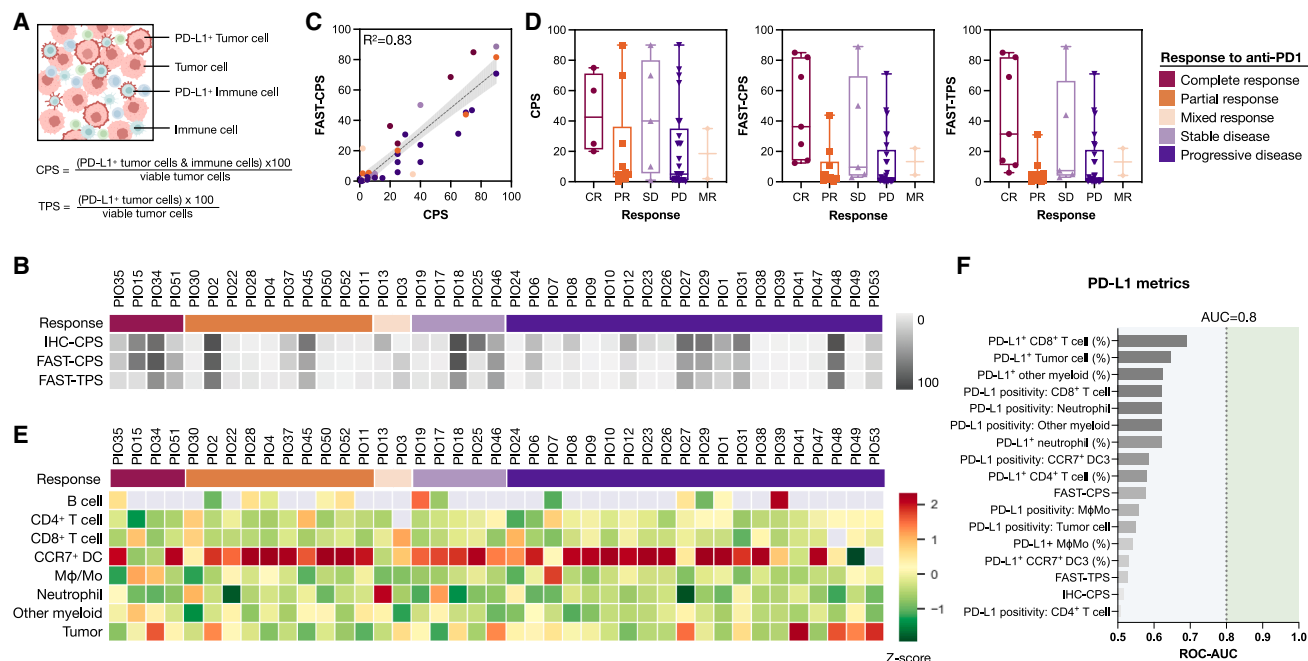


Figure 4. PD-L1 profiling alone has low predictive value for anti-PD-1 treatment response in patients with HNSCC

(A) PD-L1 expression metrics using established scoring methods.²²
(B) Heatmap of the combined positive score (CPS) scored by conventional pathology (“CPS”) or calculated by FAST multiplexed imaging (FAST-CPS), and TPS calculated by FAST imaging (FAST-TPS) across all samples. See also Figure S4 for PD-L1 signal intensity across different cell types in the TME.
(C) Correlation between manual scoring of IHC stain of PD-L1 by two pathologists (CPS) and automated scoring by FAST multiplexed immunofluorescence (FAST-CPS) (CR: complete response, maroon, $n = 4$; PR: partial response, orange, $n = 10$; MR: mixed response, light purple, $n = 2$; SD: stable disease, light purple, $n = 5$; PD: progressive disease, dark purple, $n = 20$). Dotted line shows simple linear regression with 95% confidence interval bands ($R^2 = 0.83$).
(D) Distribution of CPS, FAST-CPS, and FAST-TPS values in the different patient groups. There was no statistical significance in the values of the scores between patient groups. The boxplots indicate lower and higher quartiles and median values, with whiskers indicating the minimum and maximum values.
(E) PD-L1 expression levels across different cell types in the TME in all patient samples. Median fluorescence intensity of PD-L1 staining in different cell types was normalized per patient. Gray cells indicate no expression of PD-L1. Note that PD-L1 appears high in CCR7⁺ DCs in many patients but is not predictive in differentiating responders from non-responders.
(F) AUC of different PD-L1 expression metrics in differentiating responders from non-responders.

also higher in responders than in non-responders, as expected (Figure 5B). In addition, various subtypes of macrophages and monocytes showed increased abundance in tumor nests of responders, suggesting the need for future studies to identify their specific functional states and roles within the TME of HNSCC.

Next, we aimed to characterize two sets of potential cellular interactions: those between tumor and immune cells and those between different immune cell types. We introduced an “proximate neighbor” metric to quantify the relative prevalence of a given “neighboring” cell type within a 15 μm radius of a given “index” cell type, indicating a higher likelihood of cell-cell interaction (Figure 5C). Notably, spatial metrics with greatest fold differences between responders and non-responders pertained to the cell-cell interactions between tumor cells and CD8⁺ T cells (p value < 0.001 ; AUC 0.83; q value 0.14) and those between tumor cells and CCR7⁺ DCs (p value < 0.01 ; AUC 0.70; q value 0.62, Figures 5D and 5E). In the context of immune-immune cell interactions, the proximity of CCR7⁺ DCs to one another (p value < 0.01 ; AUC 0.70; q value 0.33) and to neutrophils (p value < 0.01 ; AUC 0.57; q value 0.55) was higher in responders to anti-PD-1 therapy than in non-responders (Figures

5F and 5G). Regarding CCR7⁺ DC-neutrophil interactions, it would be of future interest to identify the functional properties of neutrophils interacting with DCs. Recent studies have reported neutrophil subtypes with anti-tumorigenic properties.^{25,26}

DC clusters predict treatment response

Taken together, the results from spatial analyses indicated increased formation of DC niches, characterized by nests of closely situated CCR7⁺ DCs, in the tumor periphery in responders. Further analysis of CCR7⁺ DC clusters revealed that the CCR7⁺ DC clusters were predominantly located in the tumor periphery (58.9% were found inside tumor area close to the tumor boundary, and 30.8% were found in the boundary region). Of all identified CCR7⁺ DCs, 14.6% were part of clusters. These clusters most frequently consisted of five CCR7⁺ DCs in close proximity ($< 15 \mu\text{m}$), although larger aggregates were also observed. The most common immune cell types found around CCR7⁺ DC niches were macrophages/monocytes and CD4⁺ and CD8⁺ T cells (Figure 6). Upon defining clusters of CCR7⁺ DCs with greater intercellular distances (30 μm between CCR7⁺ DCs), we observed even more clusters at the tumor periphery

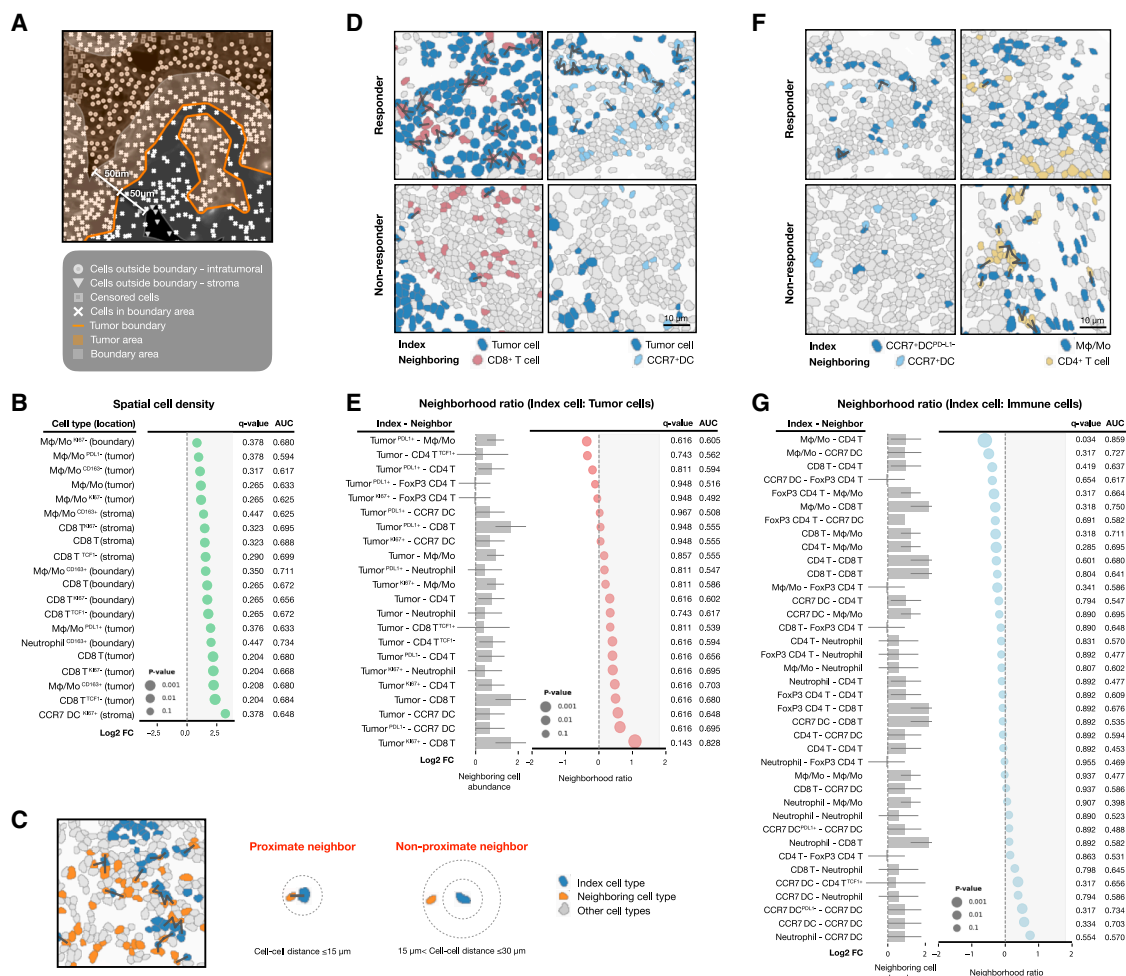


Figure 5. Spatial analysis of immune cells in the tumor microenvironment

(A) Representative image exhibiting different spatial compartments within a given FOV. Cells were classified based on their location relative to the microscopic tumor boundary (orange line): boundary (gray shade), intratumoral (orange shade), and stromal area (no shade). Areas within 50 μm from the tumor boundary were defined as boundary areas.

(B) Normalized abundance of different immune cell types located in different spatial compartments (in parentheses) was compared between responders and non-responders who received pembrolizumab monotherapy. Dot plot presents log2 fold change (FC) between responders and non-responders. For example, a high CCR7⁺ DC count in the stroma was associated with response; however, the single metric only had an AUC of 0.65 (q value 0.378) and by itself was not discriminatory.

(C) Schematic illustrating the neighborhood ratio metric, which measures the over-representation of a certain cell type among proximate neighbor cells of an index cell type (i.e., any cell in proximity to the index cell defined by a distance $<15 \mu\text{m}$). This potential cell-cell interaction was labeled with dark straight lines in the images in (C), (D), and (F).

(D–G) Neighborhood ratio metrics between tumor and immune cells (D, E) and between different immune cell types (F, G) were calculated and compared between responders and non-responders who received pembrolizumab monotherapy. The dot graphs represent the log2 fold change of neighborhood ratios between responders and non-responders, while the dot sizes indicate the p values. The bar graphs represent the log2 fold change of the total cell density of the neighboring cell types and the standard deviation of the total cell density. The q values indicate the false discovery rate calculated using the Benjamini/Hochberg method. AUC (area under the ROC curve) values are shown as a measure of the ability of metrics to differentiate responders from non-responders. Scale bar, 10 μm .

with similar composition of neighboring cells (Figure S10). Figures 6A and 6B show a representative example of a responder (PIO15) with a pronounced abundance of CCR7⁺ DC niches.

Modeling response parameters

In our investigation so far, we considered hundreds of different cellular and tissue features, grouped into three main categories:

(1) the overall density of multiple cell types and their subtypes (Figure 3), (2) the PD-L1 status of multiple cellular phenotypes, including their spatial distribution (Figures 4, S4, and S5A), and (3) the spatial organization of cells with respect to one another and tissue compartments (Figures 5 and 6). We found that certain existing metrics (PD-L1 CPS and TPS) were not predictive of immunotherapy efficacy in HNSCC, whereas several

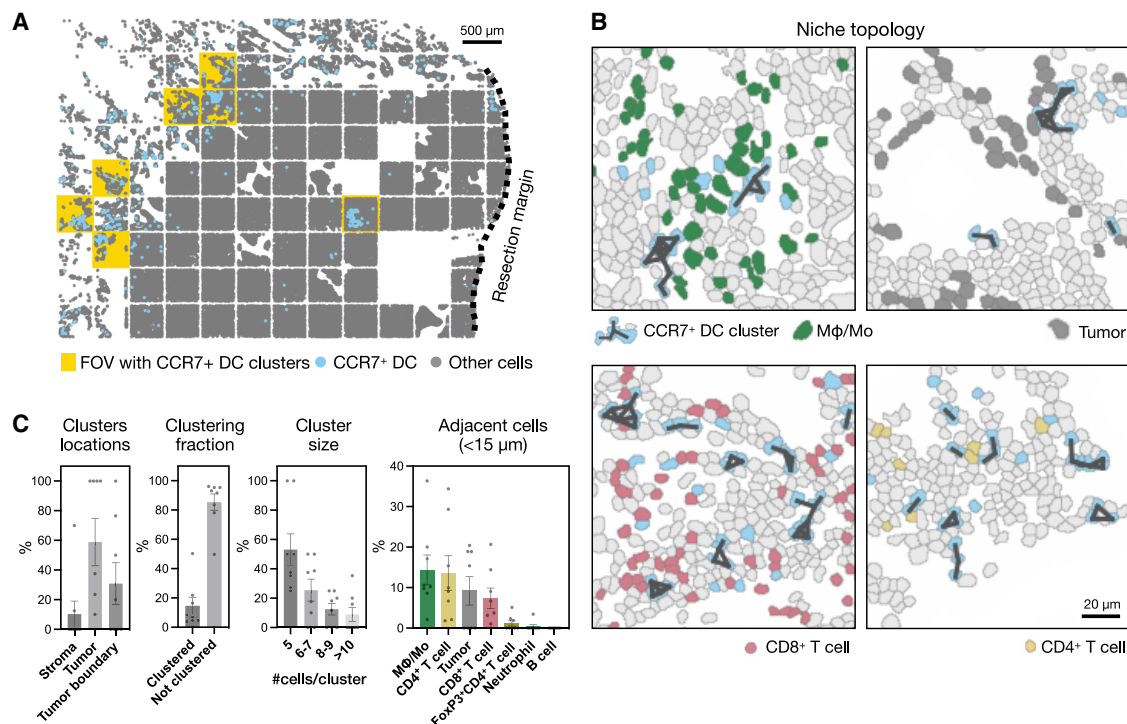


Figure 6. DC cluster profiling in a complete responder

Representative images of a complete responder (PIO15) exhibiting the clustering of CCR7⁺ DCs in the peripheral region of the tumor.

(A) Synthetic map of 205 FOVs encompassing 171,650 cells analyzed by FAST multiplexed imaging. CCR7⁺ DCs ($n = 691$) are shown as blue dots, and all other cells are shown as gray dots. CCR7⁺ DC clusters are highlighted by a yellow background and were mostly located in the periphery of the tumor tissue. The dashed line represents the surgical resection margin. Scale bar, 500 μm .

(B) Niche topology of the CCR7⁺ DC neighborhood shown in different representative FOVs. Synthetic images showing the spatial arrangements between the CCR7⁺ DC cluster (blue cells with gray connecting lines) and neighboring macrophages/monocytes (green), tumor cells (dark gray), CD8⁺ T cells (pink), and CD4⁺ T cells (beige). These clusters vary in appearance throughout the tumor tissue. Scale bar, 20 μm .

(C) Quantitative analysis of the CCR7⁺ DC cluster characteristics measuring their location, clustering fraction, size, and composition of neighboring cells. Most clusters are in the tumor periphery, are typically made up of 5 neighboring CCR7⁺ DCs (within a 15 μm radius from one another), and are often surrounded by macrophages/monocytes, CD8⁺ T cells, CD4⁺ T cells, and tumor cells. Error bars based on the standard error of the mean.

others were. This led us to explore the collective predictive capability of these metrics and to determine if the predictive performance can be improved by combined metrics. To this end, we performed multivariate regularized Cox regression analyses, leveraging the known outcomes for each patient (Figure 7). Since response and non-response to anti-PD-1 treatment is closely associated with patient survival, we utilized patient survival as a treatment outcome for the assessment of combined metrics. The overall survival was followed up to 24 months after the beginning of the therapy.

The combination of multiple tissue features significantly outperformed single features in predicting the survival outcome. This approach identified a set of MPSSs composed of different spatial cell density, cell-cell interaction, and PD-L1 metrics (see Table S5 for details on multivariate metrics). We used 5,000 FOVs from 28 patients as a training dataset and then validated the obtained metrics on another 1,200 FOVs from 10 other patients (Figure 7A, also see STAR Methods). These scores were then used to determine the p value in differentiating patients with long survival times from those with short ones, using log rank

tests following a median split applied to the whole cohort. The performance of the top 13 with p values ranging from 10^{-2} (MPSS1) to 10^{-6} (MPSS13) is shown in Figure 7B. Interestingly, many of the most predictive MPSSs identified by modeling (MPSSs 3, 4, 5, 7, and 11) contained CCR7⁺ DC metrics. For example, MPSS 11 ($p = 1.63 \times 10^{-5}$) numerically captures different aspects of a well-functioning anti-tumor immune response by combining CCR7⁺ DC density in the tumor boundary region, CCR7⁺ DC clustering, intratumoral TCF1⁺ T cell density, tumor-T cell interactions, as well as FAST-CPS to reflect the overall PD-L1 expression pattern. Figure S11 provides an overview of how well the top-ranked scores (MPSS1–13) performed on a patient-by-patient basis. Together with the overall survival analysis (Figures 7D and S7), these data show an improved performance of the new MPSS metrics compared to CPS. The fact that combinations of different metrics were more predictive than single metrics is not entirely surprising, as they reflect multiple aspects of anti-tumor immune response capabilities. In summary, the modeling allowed us to extract features that may reveal distinct drivers of immunotherapy response.

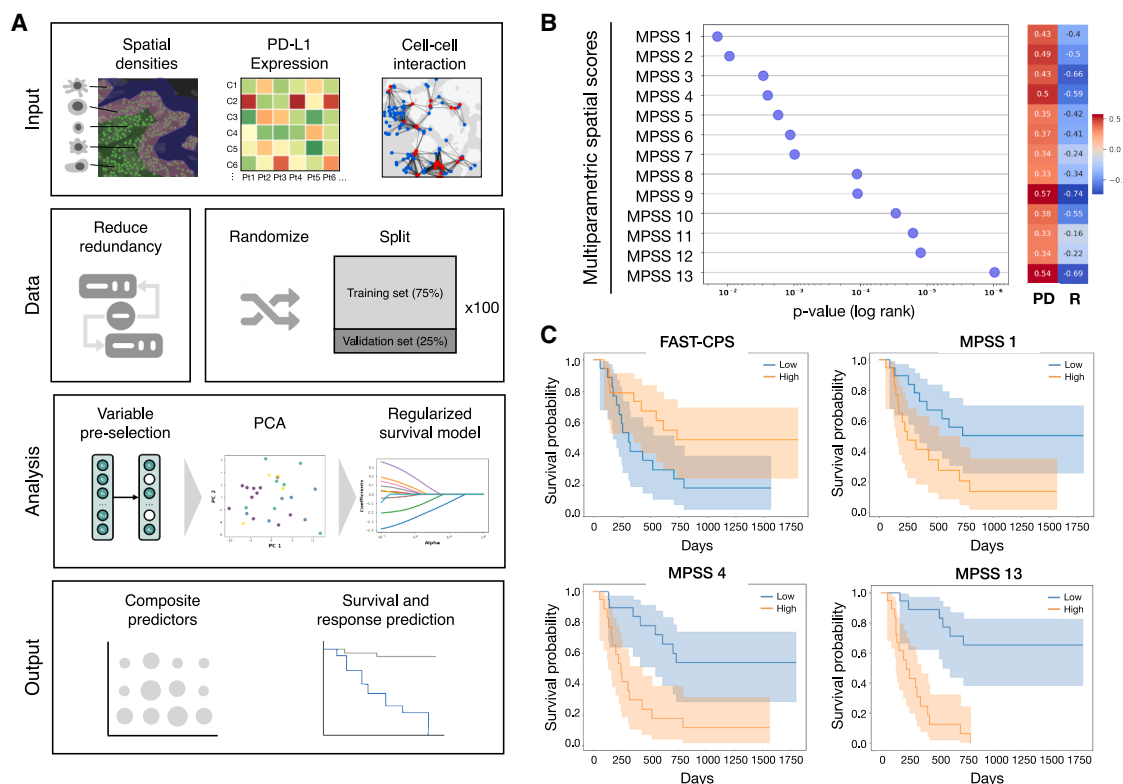


Figure 7. Modeling identifies multiparametric spatial scores that better predict the benefit of anti-PD-1 treatment in HNSCC

(A) Different classes of metrics (cell phenotype, PD-L1 expression profile, and spatial cell-cell interaction patterns) were used to model the best predictor combinations for anti-PD-1 therapy outcome in HNSCC. All patients were included in this multivariate modeling in order to use maximize the sample size. The model used a redundancy reduction step of the hundreds of metrics before randomized cross-validation. Each training dataset was constructed from ~75% of the data (5,000 FOVs from 28 patients) to derive optimized predictor scores. These were subsequently tested in a validation cohort (1,200 FOVs from 10 patients). (B) Summary of the top ranking multiparametric spatial scores (MPSSs) ranked by their ability to predict the overall survival using log rank test p values (median split). MPSSs 3, 4, 5, 7, and 11 are CCR7⁺ DC-centric scores. PD, progressive disease; R, responders. (C) Kaplan-Meier curves for the overall survival of patients grouped by representative MPSS metrics from (B) are shown. Compared to FAST-CPS (with high correlation with CPS; Figure 3), the different metrics are progressively discriminatory. See Figure S7 for the performance of MPSSs on a patient-by-patient basis.

DISCUSSION

In this study, we used a recently developed ultra-fast cyclic method (FAST) to comprehensively characterize the tumor immune landscape in patients with HNSCC to predict response to pembrolizumab. It is now routine clinical practice to use existing histopathological PD-L1 scores, such as CPS, to select patients for anti-PD-1 monotherapy. More than 85% of patients with HNSCC meet the eligibility criteria of CPS > 1, but only 10%–20% respond to anti-PD-1 therapy. Thus, there has been intense interest in developing better predictive scores and analytical systems.²⁷ The challenge to date has been to identify such metrics and test them prospectively in sufficiently large sample sets. We initially reasoned that three different approaches could be pursued to identify predictive biomarkers: (1) a more nuanced cell-based PD-L1 profiling (e.g., cell type-specific PD-L1 profiles), (2) re-examining pathways and biomarkers identified as predictive in other cancers and applying them to HNSCC (e.g., the presence of tertiary lymphoid structures [TLSs],²⁸ TCF1⁺ T cells,¹⁵ or proliferating CD8⁺ T cells²⁹), and (3) exploring emerging new

spatial biomarkers based on our current understanding of the immune response in HNSCC.⁷ Using this comprehensive multipronged approach, a new multiplex imaging method for the analysis of the tumor immune landscape at the single-cell level, and the largest spatial dataset reported to date in HNSCC, our data support the limited predictive utility of PD-L1 profiling and point to more predictive spatial features. While CPS scored by the pathologists (IHC-CPS) and FAST imaging (FAST-CPS) showed decent correlation ($R^2 = 0.83$), they failed to efficiently differentiate the responders and non-responders to pembrolizumab in our dataset (AUC 0.52 for IHC-CPS and 0.58 for FAST-CPS). Our PD-L1 profiling on different cell types in different tissue compartments of tumors showed that PD-L1 expression in myeloid cell populations can improve the AUC values, although moderately. Our single-cell spatial analysis, on the other hand, indicated pronounced formation of CCR7⁺ DC niche in responders. Furthermore, MPSSs that included CCR7⁺ DC niche metrics best predicted patient survival.

The FAST bioorthogonal method used in this study⁶ was chosen because it allows rapid cyclic imaging of tissues with

exceptionally gentle imaging conditions and specific chemical reactions. Compared to other cycling methods that require physiochemical processing (such as CyCIF,³⁰ CODEX,³¹ ImmunoSABER,³² and IBEX³³), FAST quenching mediated by click chemistry does not lead to tissue deformation, cell loss over cycles,³⁴ or antigen degradation often observed with prolonged and repeated exposure to harsh conditions.³⁵ Our previous work has demonstrated that FAST and related methods are capable of multiplexed imaging in living systems⁶ and exceptionally delicate targets³⁶ and detect 40+ markers in a cellular sample.³⁷ Moreover, the fluorescent dye conjugates are readily prepared and avert the need for complex and costly construction of nucleic acid probe library. Thus, FAST imaging of formalin-fixed paraffin-embedded (FFPE) tissue section was well suited as a rapid and translatable method of predicting response. In parallel with the use of FAST chemistry, we implemented an analytical pipeline (Figure S3) with built-in quality control steps to further automate tissue analysis. This, together with whole-slide imaging, allowed us to survey the entirety of the tissue samples rather than relying on selective hotspots.

There is now evidence from this and other studies that the predictive value of PD-L1 expression in tumor tissue may actually be quite low, despite its FDA approval as a companion diagnostic test. There are a number of possible reasons for this, including (1) the complex biology of PD-L1 and its regulation on different cell types³⁸; (2) temporal variation in expression levels in the dynamic TME^{39,40}; (3) use of different assays, antibodies, and analysis systems^{20,41}; (4) differences in time of biopsy in relation to anti-PD-1 treatment and concomitant chemo-radiation treatment effects⁴²; (5) inter-observer variability among pathologists^{21,43}; (6) other coexisting morbidities in patient population⁴⁴; and (7) intratumor heterogeneity in PD-L1 expression, which could introduce variability in PD-L1 level measurement depending on which part of the tumor tissue was analyzed. PD-L1 can be expressed on many cell types of hematopoietic origin, including DCs,⁴⁵ tumor-associated macrophages,⁴⁶ T cells,¹² endothelial cells,⁴⁷ and cancer-associated fibroblasts,⁴⁸ in addition to tumor cells. Expression of PD-L1 by these different cell types has both overlapping and non-redundant roles in immune-mediated tumor control. Many cytokines and growth factors can increase PD-L1 expression on cancer cells and in the TME. Of these, interferon-gamma (IFN- γ) produced by T cells is the most potent inducer of non-constitutive PD-L1 expression in cancer cells and in the TME.⁴⁹ For these and treatment-related reasons, PD-L1 shows considerable spatial and temporal heterogeneity when tested serially.^{50,51} A potential solution to this problem could be the temporal analysis of tumors during treatment. While this may be excessive for some primary tumors, HNSCCs lend themselves to fine-needle aspiration, potentially affording a window to capture temporal PD-L1 changes.⁵² Other than PD-L1 expression level, Eastern Cooperative Oncology Group (ECOG) performance status, a measure of a patient's functional ability at the beginning of therapy, has been suggested as a significant factor in predicting the response to anti-PD-1 immunotherapy in patients with cancer.^{53,54}

In an effort to identify alternatives to traditionally accepted biomarkers such as TIL and PD-L1 expression in the TME, we performed an analysis focused on the subset of patients who were

complete or partial responders with durable treatment response along with complete non-responders to compare. We hypothesized that pre-existing immune features can point to a future treatment response (i.e., tumors are "primed") and that those features should be more prominent in cases of extreme responses to anti-PD-1. In our spatial analysis of key tumor-immune cell and immune-immune cell interactions in tissue samples collected prior to anti-PD-1 treatment, one of the emerging metrics of response was CCR7⁺ DC clusters, which were often located at the tumor boundaries. Different DC subsets, including the recently identified tumor-associated CCR7⁺ DC subset, have been defined based on their ontogeny, location, and function.^{8,9} CCR7⁺ DCs are best defined by mRNA signatures^{8,23} and express LAMP3 and PD-L1 at high levels. The tumor-infiltrating CCR7⁺ DC state is found exclusively in tissues and does not appear to have a similar counterpart in the blood. Within tumors, CCR7⁺ DCs can produce interleukin (IL)-12 upon sensing IFN- γ released by neighboring T or natural killer (NK) cells.⁵⁵ In turn, IL-12 stimulates anti-tumor CD8⁺ T cells⁵⁵ or NK cells,⁵⁶ indicating that CCR7⁺ DCs can license full-fledged anti-tumor immunity through a molecular crosstalk involving IFN- γ and IL-12. Further, these tumor-resident CCR7⁺ DCs were shown to co-localize with PD-1⁺ CD8⁺ T cells and up-regulate stimulatory molecules that promote anti-tumor cytolytic activity following anti-PD-L1 treatment.⁵⁷ CCR7⁺ DCs may not all migrate within tumors but instead remain localized in perivascular regions and produce IL-15, which can help T cells survive within the TME.^{55,58} Thus, these blood vessel-associated and tumor-resident CCR7⁺ DCs contribute to a key T cell-associated checkpoint within tumors. Moreover, CCR7⁺ DCs are likely to be important in TLS formation, which has been shown to correlate with increased patient survival in many human tumors including HNSCC.^{59–61} DCs may provide privileged sites for local presentation of tumor antigens to T cells and activation, proliferation, and differentiation of T and B cells. As they may be present not only in mature TLSs but also in incipient ones, it is of future interest to study the role of DC clusters in the formation and function of intratumoral TLSs. Increased expression of CCR7⁺ DC marker genes in tumors has been associated with improved patient survival in non-small cell lung cancer²³ and microsatellite instability-high colorectal cancer.²² In addition, the enrichment of CCR7⁺ DCs in pretreatment biopsy samples from patients with breast cancer was implicated in the expansion of tumor-infiltrating T cells after treatment⁶² and in patients with hepatocellular carcinoma who responded to neoadjuvant anti-PD-1 therapy.⁶³

In summary, this study suggests that the formation of intratumoral CCR7⁺ DC clusters in human HNSCC supports the anti-tumor immune response induced by anti-PD-1 therapy. Given the conservation of these DCs across cancer types,⁸ and their increasingly appreciated role in anti-tumor immunity,⁹ it should be of interest to further investigate the relevance of these cells in other cancer types and treatment contexts. It is anticipated that profiling of CCR7⁺ DCs could serve as a predictive biomarker for immune checkpoint inhibitor therapy. Furthermore, as TMEs are composed of various interdependent immune players, further studies are warranted to define how CCR7⁺ DCs interact with the TME and whether they are necessary and/or sufficient to achieve durable benefit.

Limitations of the study

There are limitations to the current study. We used a predefined panel of 21 antibodies to focus on the spatial biology and the roles of previously reported major cell types and subtypes in the HNSCC TME. Further expansion of the panel to include additional cell states (e.g., macrophage subtypes and neutrophil subtypes) is subject to future studies. This study used FFPE sections, which limit the spatial analysis to 2D. With further advancement of 3D tissue imaging technique for immune and tumor markers, 3D assessment of the TME will enable a more comprehensive understanding of spatial cellular organization.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Ralph Weissleder (rweissleder@mgh.harvard.edu).

Materials availability

All unique/stable reagents generated in this study will be made available on request with a completed materials transfer agreement.

Data and code availability

- The data reported in this paper will be made available upon reasonable request from the [lead contact](#).
- Custom Python scripts used for processing are available on GitHub (https://github.com/janhoelzl/HNSCC_spatial_analysis).
- Any additional information required to reanalyze the data reported in this work paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.I.P., M.J.P., and R.W.; methodology, J.O., J.H., and J.C.T.C.; software, J.H.; formal analysis, J.O. and J.H.; investigation, J.O. and J.H.; resources, J.C.T.C. and S.I.P.; data curation, J.O. and J.H.; writing – original draft, R.W., J.O., and J.H.; writing – review and editing, all authors; visualization, J.O., J.H., and H.M.P.; supervision, R.W.; project administration, R.W.; funding acquisition, S.I.P. and R.W.

DECLARATION OF INTERESTS

R.W. is a consultant to Boston Scientific, Earli, and Accure Health, none of whom contributed to this research. R.W., M.J.P., and S.I.P. are co-inventors on several patents, all of which have been assigned to Massachusetts General Hospital and are managed through institutional oversight. M.J.P. has served as a consultant for AstraZeneca, Elstar Therapeutics, ImmuneOncia, KSQ Therapeutics, Merck, Siamab Therapeutics, Third Rock Ventures, and Tidal. S.I.P. has served as a consultant for AbbVie, AstraZeneca/MedImmune, Cue Biopharma, Fusion Pharmaceuticals, MSD/Merck, Newlink Genetics, Oncolys

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STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-Pan Cytokeratin	Biolegend	Cat# 914204; RRID: AB_2616960
Anti-P63	Abcam	Cat# ab735; RRID: AB_305870
Anti-Ki67	Abcam	Cat# ab15580; RRID: AB_443209
Anti-EGFR	R&D	Cat# AF231; RRID: AB_355220
Anti-EPCAM	R&D	Cat# AF960; RRID: AB_355745
Anti-Her2	Bio-Rad	Cat# MCA1788; RRID: AB_322791
Anti-MUC1	Biolegend	Cat# 355602; RRID: AB_2561642
Anti-CD45	Cell Signaling Technology	Cat# 47937; RRID: AB_2922773
Anti-CD3	Abcam	Cat# ab11089; RRID: AB_2889189
Anti-CD4	Abcam	Cat# ab280849; RRID: AB_3107080
Anti-CD8	Cell Signaling Technology	Cat# 85336; RRID: AB_2800052
Anti-FoxP3	Thermofisher	Cat# 14-4777-80; RRID: AB_467555
Anti-CD20	Thermofisher	Cat# 14-0202-82; RRID: AB_10734340
Anti-CD11b	Cell Signaling Technology	Cat# 23743; RRID: AB_3683724
Anti-CD68	Biolegend	Cat# 916104; RRID: AB_2616797
Anti-CD14	Cell Signaling Technology	Cat# 43878; RRID: AB_3683725
Anti-CD11c	Cell Signaling Technology	Cat# 45581SF; RRID: AB_2799286
Anti-LAMP3	Atlas Antibodies	Cat# HPA051467; RRID: AB_2681475
Anti-CD66b	Biolegend	Cat# 305102; RRID: AB_314494
Anti-PD-L1	Cell Signaling Technology	Cat# 13684; RRID: AB_2687655
Anti-PD-1	Cell Signaling Technology	Cat# 63815SF; RRID: AB_3675993
Anti-CTLA4	Abcam	Cat# ab237712; RRID: AB_2905652
Anti-TCF1	R&D	Cat# AF5596; RRID: AB_2199293
Anti-CD163	Bio-Rad	Cat# MCA1853; RRID: AB_2074540
Anti-Rabbit IgG	Thermofisher	Cat# 31210; RRID: AB_228334
Anti-Mouse IgG	Biolegend	Cat# 405301; RRID: AB_315005
Anti-Rat IgG	Abcam	Cat# ab97053; RRID: AB_10698225
Biological samples		
HNSCC FFPE sections	Participants of this study	N/A
Chemicals, peptides, and recombinant proteins		
AFDye 488 TFP Ester	Fluoroprobes	Cat# 1026-5
AFDye 555 NHS Ester	Fluoroprobes	Cat# 1166-5
AFDye 647 NHS Ester	Click Chemistry Tools	Cat# 1121-5
dTCO-PNP	Dr. Hannes Mikula	https://www.ias.tuwien.ac.at/hannesmikula
BHQ@-3 Amine	LGC Biosearch Technologies	Cat# BHQ-3001-5
Amino-dPEG4-carboxylic acids	Quanta BioDesign	Cat# 10244
Amino-dPEG6-carboxylic acids	Quanta BioDesign	Cat# 10067
N- α -Boc-N- ϵ -Fmoc-Lysine	Chem-Impex	Cat# 00037
Dimethyl Sulfoxide	Millipore Sigma	Cat# 472301-500ML
Software and algorithms		
GraphPad Prism (v.10.1.1)	GraphPad software	RRID: SCR_002798
FIJI ImageJ (v.2.1.0)	FIJI	https://fiji.sc ; RRID: SCR_002285

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Python (v.3.8.5/3.7.12)	Python software Foundation	https://www.python.org
BaSiCPy (v.1.0.0)	Peng et al. ⁶⁴	https://github.com/peng-lab/BaSiCPy
OpenCV (v.4.5.1/4.7.0)	OpenCV Team	https://pypi.org/project/opencv-python/
DeepCell (v.0.12.4)	Vanvalen lab	https://github.com/vanvalenlab/deepcell-toolbox
Pillow (v.8.0.1/9.0.1)	Pillow team	https://pillow.readthedocs.io/en/stable/
Scikit-image (v.0.18.3/0.19.2/0.19.3)	van der Walt et al. ⁶⁵	https://scikit-image.org
Alphashape (v.1.3.1)	Edelsbrunner et al. ⁶⁶	https://pypi.org/project/alphashape/
Shapely (v.2.0.1)	Gillies et al. ⁶⁷	https://shapely.readthedocs.io/en/latest/
Scikit-learn (v.1.0.2)	Pedregosa et al. ⁶⁸	https://scikit-learn.org/stable/
Statsmodels (v.0.12.0/0.13.5)	Seabold and Perktold ⁶⁹	https://www.statsmodels.org/stable/
Scikit-survival (v.0.17.2/0.22.1)	Pölsterl ⁷⁰	https://scikit-survival.readthedocs.io/en/stable/
Lifelines (v.0.27.8)	Davidson-Pilon et al. ⁷¹	https://lifelines.readthedocs.io/en/stable/
Tkinter (v.8.6.10)	Lundh et al. ⁷²	https://docs.python.org/3/library/tkinter.html
Networkx (v.2.5/2.6.3)	Hagberg et al. ⁷³	https://networkx.org
Numpy (v.1.19.2/1.21.6)	Harris et al. ⁷⁴	https://numpy.org
Pandas (v.1.2.0/1.3.5)	McKinney et al. ⁷⁵	https://pandas.pydata.org
Scipy (v.1.5.2/1.7.3)	Jones et al. ⁷⁶	https://scipy.org
Matplotlib (v.3.3.2/3.5.1/3.5.3)	Hunter ⁷⁷	https://matplotlib.org
Seaborn (v.0.11.0/0.11.2/0.12.2)	Waskom et al. ⁷⁸	https://seaborn.pydata.org
Descartes (v.1.1.0)	Descartes Labs Platform	https://pypi.org/project/descartes/
R (v.4.1.2)	R Core Team	https://www.r-project.org/
DESeq2 (v.1.34.0)	Love et al. ⁷⁹	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
BioRender	Science Suite Inc.	https://biorender.com/
Custom image processing and analysis code	This paper	https://github.com/janhoelzl/HNSCC_spatial_analysis

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

The study was performed under an institutional review board–approved protocol at the Massachusetts General Hospital and Massachusetts Eye and Ear Infirmary (IRB 2014P000559, IRB#13–416, IRB#11–024H) and is compliant with the Declaration of Helsinki and Health Insurance Portability and Accountability Act regulations. The eligibility criteria included adult patients with a histologic diagnosis of HNSCC and who had surgical excisions ($n = 38$) or core biopsies (>30 FOVs; $n = 10$) of their lesions prior to treatment with anti-PD-1 immunotherapy. We selected for patients who had not received any prior systemic treatment before the tissue collection. We further required more than 2 cycles of immunotherapy and available tissue with cancer cells present. A total of 48 patients samples were collected and analyzed. The age, sex, and additional clinically relevant information is provided in [Table S3](#) and [Figure S2](#).

HPV status of oropharyngeal tumors was determined based on p16 expression by immunohistochemistry (IHC) and/or by HPV 16/18 *in situ* hybridization (ISH) and/or HPV PCR. Nasopharyngeal squamous cell carcinoma cases were tested for EBV by EBER ISH. Clinical demographic information, such as age, sex, smoking, and others, were extracted from electronic medical records retrospectively. Non-smokers were defined as having a smoking history of <10 pack years. Response was defined as the best response achieved within 6 months after starting immunotherapy, based on RECIST-measurable disease if available, and/or based on clinical assessment. “Non-responders” were patients with progressive disease (PD), stable disease (SD) at around 6 months after starting immunotherapy, or mixed response (MR), whereas “responders” are those with a partial (PR) or complete response (CR). Cases that did not qualify as CR/PR or PD/SD by clinical/radiological assessment were assigned as MR.

METHOD DETAILS

Overall design

The initial goal of the study was to determine how well PD-L1 profiling in tissue section predicts subsequent clinical response to anti-PD-1 therapy in HNSCC. The secondary goal was to determine whether immune subset profiling could improve on global CPS measurements and whether there existed any other spatial tissue metrics indicative of future response. We designed a protein panel of 21

markers (Table S1), so that most of major immune cell types and some of their subtypes in the TME could be classified spatially. We then obtained multiple antibody clones in carrier-free solutions to conjugate FAST probes by amine-reactive chemical groups (Figure S1A). The FAST-labeled antibodies were then tested in normal tonsil sections and HNSCC tissue microarrays (Figure 1B). IgG isotypes were used for negative controls. Once we had identified the optimal FAST-antibody conjugate for a given molecular target, we performed additional immunohistochemistry to validate that the FAST-labeled antibodies could appropriately detect molecular markers compared to gold standard histology (Figures S1B and S1C). Once appropriately validated, we used the FAST-labeled antibody panel to develop a multiplexed tissue imaging pipeline. The overall study design is summarized in Figure S3. The sections below detail the different steps.

Synthesis of FAST probes

FAST probes are built to link fluorochromes and antibodies with an embedded *trans*-cyclooctenes (TCO) for clicking with a tetrazine (Tz)-quencher (Figure S1). FAST probes were synthesized at large scale as described in detail in our previous studies¹⁰ and stored as the carboxylic acids, to be activated *in situ* for antibody labeling by NHS-ester amine reaction. The dTCO-PEG6-CO₂H blocking reagent was synthesized from dTCO-PNP and amino-dPEG6-CO₂H and characterized by LCMS. rTCO-PNP and dTCO-PNP were a generous gift of Dr. Hannes Mikula (TU Wien, Austria). All other reagents were obtained from commercial sources at the highest grade available and used without further purification. Fluorophores were purchased from Click Chemistry Tools or Fluoroprobes. BHQ-3 Amine was purchased from LGC Biosearch Technologies. N- α -Boc-N- ϵ -Fmoc-Lysine was purchased from Chem-Impex. Amino-dPEGn-carboxylic acids ($n = 4,6$) were obtained from Quanta BioDesign. Dry solvents and coupling reagents were obtained from Sigma Aldrich.

FAST antibody conjugation

Carrier-free antibodies were purchased (Table S2) and modified with FAST probes as follows. Antibodies were exchanged into bicarbonate buffer (pH 8.4) using a 40k Zeba column (Thermo Fisher). Antibodies were then incubated with an appropriate molar excess of the Dye-TCO-NHS molecule (FAST probe) with 10% DMSO for 25 min at room temperature protected from light. The molar excess of FAST probe was determined based on the initial concentration of the carrier-free antibody. For removal of unreacted FAST probes, the conjugation reaction was loaded onto a 40k Zeba column equilibrated with PBS. The degree of labeling (DOL) of FAST antibodies were determined by measuring the absorbance spectrum using Nanodrop 1000 (Thermo Scientific). The known extinction coefficients of the dye, IgG antibody, and correction factor for the dye absorbance at 280nm were applied for the calculation of DOL. The FAST-labeled antibodies were stored in the dark at 4°C in PBS until usage. All antibody conjugates were validated using human tonsil tissue sections prior to FAST imaging.

FFPE sample preparation

Tissues used for FAST imaging were cut from archived formalin-fixed paraffin-embedded (FFPE) tissue blocks at 5 μ m thickness and stored at -80°C before imaging. FFPE tissue sections were baked at 55°C overnight on a slide warmer (Newcomer Supply). Then, tissue sections were de-paraffinized in xylene, rehydrated in a series of graded alcohols, and washed in water. For antigen retrieval, tissue sections were placed in Tris-based antigen unmasking solution (Vector Laboratories) in a de-cloaking chamber (Biocare Medical), boiled at 110°C under full pressure for 1 min, and slowly cooled to 80°C . Finally, sections were incubated in water (5 min, 2 times) and PBS (5 min). Following de-paraffinization and antigen retrieval, FFPE tissue sections were blocked with Intercept buffer (LI-COR Biosciences) for 30–60 min and permeabilized with 0.1% Triton X-100 for 30 min. Then, a microfluidic tape device⁸⁰ was installed around the tissue on the glass slide to allow efficient fluid exchange under coverslip for repeated immunostaining for FAST imaging.

FAST imaging

Immunostaining for FAST imaging was performed in accordance with typical immunofluorescence protocols. For additional validation of the FAST-labeled antibodies, consecutive sections of human tonsil tumor were prepared as described above, stained with the same primary antibodies used for FAST imaging, and developed for IHC with Dako EnVision FLEX kit (Agilent). The FAST-conjugated antibodies were diluted to 1–5 $\mu\text{g/mL}$ in Intercept buffer for staining. In each cycle, tissue sections were incubated with antibodies for 2 h for surface markers and overnight for intracellular or nuclear markers, and then washed with PBS 3 times, each 10 min in duration. Following image acquisition of the tumor area, tissue sections were incubated with 10 μM Tz-BHQ for 30 min in PBS-bicarbonate buffer (pH 9) to quench the fluorescent signal. Free Tz-BHQ was removed by three washes with PBS, and the tissues were imaged again in the same set of FOVs to record the quenched signal. Before antibody staining of the subsequent cycle, tissue sections were briefly incubated in a solution of 10 μM dTCO-PEG6-CO₂H in order to block any residual Tz-BHQ3 from reacting with FAST antibodies of the next cycle. The same staining, imaging, and quenching cycle was repeated until all of the target proteins were imaged. A tonsil section was stained and imaged following the same protocol as above in each batch of imaging samples as a positive control for the immunofluorescence of target protein markers.

Each tissue section was scanned by slide scan module in Metamorph with 10% overlap, using Olympus BX-63 upright epifluorescence microscope (20x objective resulting in a FOV of $561.6 \times 561.6 \mu\text{m}$). The light source was an X-Cite XYLIS system (Excelitas, Mississauga, ON, Canada), and the imaging camera was an ORCA-Fusion (Hamamatsu). The following Semrock filters were employed for different imaging channels: Hoechst: OSF3-DAPI-1160B-Z (Ex: 387/11; Em: 447/60nm); AF488: OSF3-GFP-3035D-Z

(Ex: 472/30; Em: 520/35 nm); AF555: OSF3-CY3-4040C-Z (Ex: 531/40; Em: 593/40nm); AF647: OSF3-CY5-4040B-Z (Ex: 628/40; Em: 692/40 nm).

PD-L1 immunohistochemistry of tissue sections

The MGH PD-L1 immunohistochemistry is a laboratory-developed test (LDT) with clone E1L3N (Cell Signaling Technologies). It is generally understood that the performance of the E1L3N clone is very similar to most PD-L1 antibodies/companion diagnostics including Dako 22C3 and others. Expression of PD-L1 for each sample was scored by head and neck pathologists and the combined positive score (CPS) was determined as previously described.¹⁹

QUANTIFICATION AND STATISTICAL ANALYSIS

Pipeline of multidimensional image processing and quality control

Images analysis was done using Python and FIJI, as summarized in [Figure S3](#), and is described in detail below. Briefly, all raw images were corrected for uneven illumination, and images from each cycle and quench were aligned to adjust for pixel shifts and rotation. After alignment, quenched images were subtracted from stained images of subsequent cycle for background subtraction. For cycle 1, images of the unstained sample ('cycle 0') were subtracted. The nucleic and cytoplasmic areas of single cells were segmented, and the fluorescent intensity of all protein markers in each identified cell was measured in the aligned quench-subtracted images. Cell type classification was done according to well-established marker assignments, as previously done with conventional IHC and flow cytometry. Raw single-cell data from all FOVs that passed all the QC steps were analyzed for (1) tumor/immune cell composition, (2) PD-L1 status, and (3) spatial analysis. Custom Python scripts used for processing are available on GitHub (https://github.com/janhoelzl/HNSCC_spatial_analysis).

Illumination correction

To correct for both multiplicative (uneven illumination) and additive (thermal noise, camera offset) unwanted influences in the acquired images, the first step in our analysis pipeline used BaSiCpy, a retrospective shading correction method based on a sparse and low-rank decomposition.⁶⁴ The correction was done separately for each imaging cycle and channel as follows: For every sample batch, images from all samples with >60 fields of view (FOV) were collected. Flatfield (multiplicative) and darkfield (additive) correction matrices were then created separately for each of them with BaSiC, followed by the calculation of pairwise mutual information for the flatfield matrices using a custom function. An undirected graph was generated by thresholding the matrix of pairwise MIs at a value of 0.1 after which the largest clique (or the one with the highest mean MI in case of a tie) was extracted in order to exclude outlier samples where the correction matrices created by BaSiC might not be robust. The mean of flatfield and darkfield correction matrices from the samples within this clique was then used to correct all samples within the batch. This approach was chosen because after 60 images, improvements in BaSiC's performance are limited, and we observed that including an averaging step for each batch improved the robustness of illumination correction.

Alignment and quench subtraction

Following illumination correction, raw DAPI images from all subsequent cycles and quenches were matched to their corresponding cycle 0 images using the Python implementation of OpenCV (<https://pypi.org/project/opencv-python/>). The resulting shift and rotation values were used to align all images for a given FOV to cycle 0 and crop them so that only tissue areas present in all cycles and quenches was kept. To avoid double-counting of cells due to overlap between FOVs caused by slide scanning with 10% overlap, a minimum of 10% (173 pixels) was cropped from the top and left side of each image. FOVs without tissue were disregarded. FOVs were classified as misaligned if one or more of the MI values between the cycle 0 DAPI image and all subsequent aligned DAPI images were less than 0.06. For all FOVs passing this alignment quality control, quench subtracted images were then created and saved. In addition, the mutual information between a given cycle and a previous one was minimized, based on the fact that in any individual channel, distinct staining patterns can be expected in each cycle. After setting any negative valued pixels to 0, the resulting optimized quench subtracted images were used for downstream measurements.

Cell segmentation

For nuclear segmentation, DAPI images from the last image cycle were used. For cytoplasmic segmentation, we used aligned and quench-subtracted images of CD45 and panCK. These were first normalized to 16 bit range for each FOV, then, a maximum projection of the two images was generated. This projection was combined with the DAPI image into a 3days array and used to generate nuclear and cytoplasmic masks with Mesmer (resolution setting: 0.5), a deep learning-based segmentation tool from the DeepCell toolkit (<https://github.com/vanvalenlab/deepcell-toolbox>). Basic metrics like mean fluorescence intensity, x and y coordinates, area, and others were then extracted for each cell using Scikit-image. Additional metrics (described in detail below) were calculated using custom functions. Mesmer-generated cytoplasmic masks were used to extract data for cytoplasm and membrane-localized proteins, and nuclear masks were used for nuclear proteins.

Artifact detection and removal

Following the measurements, autofluorescence signal was filtered based on the mean fluorescence intensity in cycle 0 (unstained) by setting thresholds for autofluorescence in each channel. We occasionally observed unspecific signal from evenly shaped artifacts in the raw images, which we attribute to illumination anomalies and/or impurities on the coverslip. As these artifacts can interfere with proper cell segmentation, thresholding and cell type annotation downstream, they were identified using a custom function to exclude

cells localized within them. Briefly, it compares the cycle - quench image pair used to generate a given quench-subtracted image by applying a Discrete Fourier Transform, subtracting the images in frequency space, and identifying possibly affected FOVs as the ones with a high ratio of low frequencies to high frequencies added after staining. The areas containing those pronounced new low frequencies are then identified, and the procedure is repeated locally, leading to a final classification of the local tissue region as artifact-affected or non-artifact-affected. All cells within the artifact-affected tissue areas were excluded from downstream analysis. This approach proved to be the most suitable for the removal of these larger and evenly shaped artifacts as strong cellular staining manifests itself in a strong mid - high frequency signal whilst the aforementioned artifacts produce a very low intensity signal. Therefore, this Fourier based approach is perfectly suited for the detection of this specific type of artifact. Lastly, it was empirically determined to exclude objects with an area ≤ 400 pixel² from further analysis. This was done as it corresponds to a cell diameter of roughly 7 μm (assuming perfectly circular cells). We expect the large majority of cells of interest to be larger than this and therefore assumed smaller detected objects to be mostly artifacts and mis-segmented cells.

Cell phenotyping and measurement

In order to deal with the complexity of correctly assigning the cell identities in whole-slide immunofluorescence images of FFPE tissue sections, a novel phenotyping strategy utilizing three custom metrics was used for each marker. The three metrics include the measurement of the interquartile mean (IQ_mean: mean fluorescence intensity of the middle 90% of pixels, excluding the top and bottom 5%; i.e., 0.05–0.95, of each cell), staining skewness (Skewness_of_stain: The distance between centroid and intensity weighted centroid divided by $\sqrt{\frac{\text{cell area}}{\pi}}$ to normalize for cell size), and entropy of a cell's staining pattern (Prop_50: the proportion of pixels necessary to account for 50% of the cell's total fluorescence intensity when a cumulative sum was measured from ranked pixels in descending order). IQ_mean was used as our main metric of fluorescence intensity to minimize influences by outlier pixels. Skewness_of_stain represents the degree of “off-centerness” of the staining and was used to avoid false positives caused by overlapping cells in the densely packed tumor tissue. Prop_50 was used to avoid false positive caused by autofluorescent tissue areas. (see “measurements” in Figure S3A).

The three metrics above were used to assign either “positive” or “negative” status to a given cell for each cytoplasmic or membrane immune cell marker by creating a sample-wise threshold for each and applying these to all analyzed cells. First, a threshold suggestion was automatically generated with the following approach: a function of the form $y = b(1 - e^{-kx})$ was fit to the data for each sample where y is a scaled version of Prop_50 and x a scaled version of IQ_mean of a given cell. Fitting above function and setting the IQ_mean threshold at the “elbow” of the graph, which we identified through the first derivative of the function, resulted in good separation of negative and positive cells. Then the threshold was set in two steps: i) to identify a group of high confidence positive cells used to generate a Skewness_of_stain threshold, followed by exclusion of all cells below this threshold, and ii) a re-fitting of the function and the calculation of final IQ_mean and Prop_50 thresholds. Skewness_of_stain and Prop_50 thresholds were both determined by applying the full-width at half maximum (FWHM) method to a subset of high confidence positive cells (high fluorescence intensity). The location and FWHM of the population peak were determined by applying a kernel density estimation (KDE) to the respective histograms followed by peak identification and FWHM measurement. The thresholds were then set $\text{scale_factor} \times \text{FWHM}$ away from the population peak. This method was also used to generate automatic thresholds for nuclear and/or tumor proteins. Here, log2-transformed IQ_mean values were used for the KDE, based on the assumption that fluorescence intensity is approximately log-normally distributed. Following the generation of all pre-thresholds, we used a custom graphical user interface (GUI) to manually check their validity and make manual adjustments if necessary before applying the final thresholds and assigning a list of positive markers to each cell.

Phenotype assignment was performed as the following. First, the fluorescent signal of all tumor markers (PanCK, p63, Quad) and CD45 and Ki67 were thresholded to classify individual cells as tumor or immune cells (or undefined if negative for all lineage markers). Then, all undefined and immune cells were subjected to the thresholding of the remaining immune markers and were assigned precise identities based on their list of positive markers. If more than one assignment was possible, conflicts were resolved by comparing the percentile among all cells positive for a given marker for all non-overlapping cytoplasmic markers between groups. Cells with more than 4 possible phenotypes or otherwise unresolvable conflicts were assigned “unresolvable_conflict” status, those without any matching cell type were called “undefined”. The markers used for the assignment of each cell type are summarized in Figure 2C. Finally, CD163 was thresholded using only myeloid cell populations, FoxP3 and TCF1 using only T cell populations, and PD-L1 using all cells. These markers, along with Ki67, were used for subtyping all previously phenotyped cells by assigning “positive,” “negative,” or “not assessed” status.

In order to define tumor area in macro-scale, the adjacent tissue sections of each patient samples used for FAST imaging was H&E stained according to standard protocol. The slides were then reviewed by pathologists specialized in head and neck cancers to circle the tissue area consisting of tumor. These annotations were manually recreated on stitched whole-slide scans of the DAPI channel using Fiji. Next, FOVs far outside of the tumor area were filtered for downstream analysis (spatial analysis, general statistics). FOVs with less than 25 cells were also excluded from further analysis.

Neighborhood analysis

The calculation of the type 1 neighborhood ratio $A \times B$ of a pair of cell types A and B was done as follows. All cells within 15 μm of a cell of type A were assigned to cell type A's immediate neighborhood (“IN^A”), all cells within a distance between 15 μm and 30 μm to their nearest neighbor of type A were assigned to cell type A's wider neighborhood (“WN^A”). Then, the cell frequency of cell type B was

calculated both in IN^A and in WN^A across the whole analyzed sample area. The neighborhood ratio was finally defined as cell type B's frequency in IN^A divided by its frequency in WN^A . Censoring of low abundance samples and zero-handling was done analogous to the type 2 neighborhood ratio. Additionally, samples with 0 type B cells in IN^A and WN^A combined were also censored. The following equation shows the calculation of neighborhood ratio type 1:

$$\text{Neighborhood ratio 1} = \frac{\text{Number of Cell type B within } IN_A / \text{Number of other cell types within } IN_A}{\text{Number of Cell type B within } WN_A / \text{Number of other cell types within } WN_A}$$

The calculation of the type 2 neighborhood ratio $A \times B$ of a pair of cell types A and B was done as follows. All cells within 30 μm of a cell of type A were assigned to cell type A's neighborhood (" N^A "), all others were defined as being outside of cell type A's neighborhood (" NN^A "). Then, the cell frequency of cell type B was calculated both in N^A and in NN^A across the whole analyzed sample area. The neighborhood ratio was finally defined as cell type B's frequency in N^A divided by its frequency in NN^A . If less than 10 cells of type B were present in a given sample, the ratio was not calculated due to the low confidence in the resulting estimate. If the number of type B cells within N^A was 0, the minimum of the following two values was used. 1: The minimum neighborhood ratio of all samples with no 0 present in the calculation; 2: The neighborhood ratio for that sample after replacing the 0 by a pseudo-count of 1. If the number of type B cells within NN^A was 0, an analogous approach was used while using the maximum instead of the minimum, respectively. The following equation shows neighborhood ratio type2 calculation:

$$\text{Neighborhood ratio 2} = \frac{\text{Number of Cell type B within } N_A / \text{Number of other cell types within } N_A}{\text{Number of Cell type B within } NN_A / \text{Number of other cell types within } NN_A}$$

Spatial abundance analysis

To determine how immune cells are distributed at the macro-scale, the tumor boundary was identified in each sample by a pathologist. This allowed us to assign whether a given FOV was inside the tumor, outside the tumor in the normal resident tissue, or whether it captured the tumor boundary region.

To determine how immune cells are distributed at the micro-scale, we performed a boundary analysis. Tumor cell clusters (>7 cells) were identified by thresholding a distance matrix of all tumor cells at 20 μm and extracting all connected clusters of sufficient size from the resulting undirected graph. An alpha shape was then created around each tumor cell cluster using the Alphashape python package (alpha: 0.01) and buffered by 7 μm . The union of these polygons was then defined as tumor area, and the complement as stroma. Tumor boundary lines without adjacent stroma were excluded. The remaining valid boundary lines were then buffered by 50 μm , resulting in a set of polygons in each FOV. For further quality control, it was asserted that each boundary polygon contained at least 3 cells in the part not overlapping with the corresponding tumor area. The union of the intersections of these boundary polygons with the tumor polygons was defined as the "boundary inside" in each FOV. The complement of the boundary inside within the full union of boundary polygons was defined as the "boundary outside". The Median distance to the tumor boundary was measured using cells within the "boundary outside", whose distance to the next valid tumor boundary line was equal to the minimum distance to any tumor area and less than the distance to the outer FOV border.

Cell abundances in form of raw counts were first spatially normalized using the following approach: For each spatial compartment (tumor, stroma, boundary, total), the geometric mean of its area was calculated across all samples. A spatial normalization factor was then calculated for each sample and compartment by dividing this geometric mean by the measured area in that sample. Compartment-wise cell counts were then multiplied by this factor before further statistical processing to account for differences in imaged sample area.

Hot vs. cold tumor analysis

We used established metrics to define "hot" and "cold" tumors. The "Immune-active" or "hot" phenotype is characterized by lymphocyte infiltration in the tumor parenchyma, with the immune cells positioned in proximity to tumor cells.¹² The "Immune-desert" or "cold" phenotype is characterized by a lack of lymphocyte in the tumor parenchyma or the periphery.¹² The "Immune-excluded" phenotype is characterized by abundant immune cells in the tumor periphery without penetration into the parenchyma.¹³ Specifically, samples with $CD8^+$ T cell infiltration of <500 cells per mm^2 were classified as "immune-desert," and among the samples with ≥ 500 $CD8^+$ T cells per mm^2 , samples with tumor-to-stromal ratio of $CD8^+$ T cell infiltration ≥ 0.5 were assigned as an "inflamed" or "hot" phenotype. Samples with tumor to stromal ratio of $CD8^+$ T cell infiltration <0.5 were classified as "immune-excluded" tumors.¹²

Modeling

Multivariate modeling was used to explore the use of combination biomarkers to predict response to anti-PD1 therapy and using overall survival as the endpoint (dependent variable). For multivariate modeling, data from all patients were used. The following variables were used as input (Figure 7): Spatially normalized cell abundance (all compartments), spatial PD-L1 expression ratios and neighborhood ratios (Figure 5). Ratio-based variables were used in their log-transformed form and cell abundances were rlog transformed. To compare coefficients, all values were standardized by Z - scoring. Subsequently, the feature space was reduced by calculating and thresholding correlation matrices across all samples using both Pearson (threshold = 0.9) and Spearman (threshold = 0.95) correlation coefficients. This resulted in thresholded correlation matrices (data matrix with 0 indicating no connection and 1 indicating an edge between two data points) where the two variables were treated as "connected" whenever either Pearson R was >0.9 or Spearman rho was >0.95. The variable with the highest mean log2-transformed cell count across the involved cell types was

extracted. After removing possible duplicates, the remaining feature set, along with standardized PD-L1 positivity fractions and pathological scores (Figure 4), was used for downstream modeling (total $n = 880$).

The data was randomly split into training (~75%) and test sets (~25%) stratified by treatment response (3 groups: response, progressive disease, mixed/stable disease). Three additional patients were excluded from modeling due to their low FOV counts, resulting in a total of 38 samples used. We applied a soft stratification by follow-up time by excluding splits that resulted in the test set containing no sample with <200 or >1000 days of follow-up. This process was repeated to create valid 100 train-test splits. Predictive modeling was then performed as follows. Briefly, the concordance index (C-index) was calculated for each variable to extract the 100 features with the most univariate predictive power regarding the target variable (overall survival). These variables were then used as input for Principal Component Analysis (PCA). After determining the number of principal components (PCs) necessary to explain 70% of the total variance cumulatively, the respective PCs were extracted and used for downstream survival modeling. This step was introduced to minimize overfitting in the final model. In addition to these principal components, the treatment group (pembrolizumab monotherapy vs. pembrolizumab + chemotherapy) was also added as a categorical variable to each model. A multivariate Cox Proportional Hazards model (CPH regression) was then fit to the training set using Elastic Net regularization. Both the L1/L2 ratio (alpha) and the minimal shrinkage factor lambda were automatically chosen by grid search using nested randomized cross validation within the training set ($n = 50$, 70/30 splits, alpha range = {0.1, 100}, lambda range = {0.01, 1}) with C-index as the scoring metric.

Estimates of model performance in the test set across all random splits were generated by calculating the C-index and a mean cumulative time-dependent ROC-AUC over the interval of days 200–1000 after beginning of immunotherapy in each fold. Additionally, we calculated regular AUC regarding therapeutic response for all samples within the test set that could be assigned to either of the groups “R = response to pembrolizumab monotherapy” or “PD = progressive disease”.

In order to generate a global variable ranking, for each model, the dot product of the transposed matrix of PCA loadings (L_{PCA}) and the vector containing their respective CPH model coefficients (C_{embed}) was first calculated to produce coefficients related to original variables:

$$C_{orig} = L_{PCA}^T \cdot C_{embed}$$

The entries of C_{orig} were then saved for each variable in every fold (with zeros for all variables not included in a certain fold’s model) to finally create 100 x 1 vectors containing all coefficients for each individual original variable. Absolute means of these vectors were then used to rank variables as shown in Figure 7.

In order to obtain top ranking composite scores, the 5 highest ranking variables were extracted along with their mean coefficients and the final score for each patient calculated as:

$$\sum_{i=1}^5 Var_i \times C_{mean,i}$$

where Var_i is the measured value for the i ’th highest ranked variable and $C_{mean,i}$ is the mean model coefficient for that variable. Manually curated scores (MPSS 1–12) were calculated analogously after selecting appropriate variables by using top hits from therapeutic response analysis and multivariate survival modeling, as well as prior biological knowledge. The exact constitution of all multiparametric spatial scores (MPSS), along with the coefficients used for calculation can be found in Table S5.

Statistical analysis

Statsmodels package in Python and GraphPad Prism were used for statistical analysis. More details on statistics are provided in the modeling section.